Stan Replication of McGilchrist & Aisbett's (1991) "Regression with Frailty in Survival Analysis"

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Abstract

McGilchrist & Aisbett (1991) provide a framework that stabilizes estimation of the risk variable coefficients and frailties by allowing frailties to share a common log-normal distribution. In survival analysis, frailty is an important variable to estimate risk amongst patients, for if nothing else, it could prompt further research into patients in the study showing high frailty. When finding the maximum likelihood estimates derived from Cox's (1975) partial likelihood including frailties, the Newton-Raphson algorithm can have convergence issues due to a dense observed information matrix. McGilchrist & Aisbett (1991) provide a solution by introducing a prior log-normal distribution on frailties, resulting in a penalized partial likelihood. The following replication in Stan follows the estimation method of McGilchrist & Aisbett (1991) to perform a MAP Newton-Raphson maximization of the log posterior. Risk coefficients and frailty estimates are replicated with relative accuracy, and Bayesian NUTS sampling results are also included for comparison.

1 Introduction

In survival analysis, a frailty term in the hazard function builds on the standard Cox (1972) proportional hazards model, and captures individual-level risk that chosen risk variables do not explain. In the case of measuring repeated occurrences, frailty can be considered a common factor for recurrences for each individual. In this project we will replicate the paper "Regression with Frailty in Survival Analysis" (McGilchrist & Aisbett, 1991) in a Bayesian framework using Stan. The paper analyses two separate times to infection in 38 dialysis patients. Their statistical application estimates the effects of age, sex and disease-type on patients' time-to-infection, as well as frailties which can be intuitively thought of as a measure for differences between predicted and observed recurrence times. The problem addressed by McGilchrist & Aisbett (1991) is one of estimation instability when maximizing a Cox (1975) partial likelihood with frailties. With no analytical solution for the maximum likelihood estimates, Their approach implements a random frailty term as a shrinkage estimate to the Cox (1975) partial likelihood. Their innovative method is replicated in the following report, with a main focus on maximum a posteriori (MAP) estimation, and a secondary route with Bayesian MCMC NUTS sampling.

2 Data

The dataset used by McGilchrist & Aisbett (1991) provides the time until infection from kidney catheter insertion for patients with portable dialysis equipment. With two recurrences recorded for each patient, there are 76 observations in total for the 38 patients. Every observation consists of the time until infection (response), the patient id, and the patient's age, gender and type of disease at time of infection. Censoring is registered when the catheter is removed for any other reason.

The `kidney` dataset is available through the `survival` package in R, with the per patient frailty estimate from the original paper included.

3 Models and Methods

3.1 Original Model

McGilchrist & Aisbett (1991) utilize Cox's (1972) proportional hazards model (with individual frailty term, Z_i) to model recurrence times.

$$h(t_n, \mathbf{x}_{in}, \boldsymbol{\beta}) = Z_i \lambda(t_n) \exp(\mathbf{x}'_{in}\boldsymbol{\beta}) \tag{1}$$

where $\lambda(t_n)$ denotes the baseline hazard function, and $\exp(\mathbf{x}'_{in}\boldsymbol{\beta})$ denotes the risk multiplier, a function of a linear relationship between risk variables $\mathbf{x}$ and respective coefficients $\boldsymbol{\beta}$. The subscripts i and n denote patient i of total M (38), and recurrence n of total N (76), respectively.

Cox's (1975) partial likelihood is derived using (1), shown in the appendix. The log of the partial likelihood forms the basis for estimation in the paper. In the following, D_{in} denotes a dummy variable that equals 1 if a recurrence occurs to patient i at time point t_n, and 0 if not. It is important to note that all N time points are ordered smallest to largest and assumed to be distinct, so that $D_{in} = 1$ for only one individual at every time point t_n.

$$l = \sum_{i=1}^{M} \sum_{n=1}^{N} D_{in} \left[\log(Z_i b_{in}(\boldsymbol{\beta}, \mathbf{Z}))\right] \tag{2}$$

Where: $b_{in}(\boldsymbol{\beta}, \mathbf{Z}) = \dfrac{\sum_{j=r_{in}}^{m_i} \exp(\mathbf{x}'_{ij}\boldsymbol{\beta})}{\sum_{i=1}^{M}\left(Z_i \sum_{p=r_{in}}^{m_i} \exp(\mathbf{x}'_{ip}\boldsymbol{\beta})\right)}$

$b_{in}(\boldsymbol{\beta}, \boldsymbol{Z})$ is a measure of a person's total risk at time point t_n divided by the total risk set of everyone remaining, consisting of patients with at least one recurrence time at or after time point t_n. This can also be viewed as a measure of probability that an infection occurs to person i at t_n. m_i is the total number of recurrences recorded for the participant (2 for all in this study), and r_{in} the number of recurrences occurring to the participant before t_n plus one.

The first and second derivative of (2) yield the score function and hessian matrix. With no analytical solution for the maximum likelihood estimates, the authors use a Newton-Raphson algorithm to produce MLE estimates for frailties, $\hat{\mathbf{Z}}$, and risk variable coefficients $\hat{\boldsymbol{\beta}}$.

To provide stability to this estimation procedure, which can struggle with inverting the negative hessian matrix when there is high variance between patients' risk variables (leading to highly varied frailties), McGilchrist & Aisbett (1991) introduce a log-normal prior to the frailties. Using l to represent the joint log-likelihood of both $\boldsymbol{\beta}$ and $\mathbf{Z}$, we can view its conditional form as a penalized partial log-likelihood.

$$p(\boldsymbol{\beta}, \mathbf{Z}|\mathbf{t}) \propto p(\mathbf{t}|\boldsymbol{\beta}, \mathbf{Z})p(\mathbf{Z})p(\boldsymbol{\beta})$$
$$l \propto l_1 + l_2 + 0$$

Under a non-informative flat prior ($p(\boldsymbol{\beta}) \propto 1$) on $\boldsymbol{\beta}$, its log likelihood = 0. The maximization of $l_1 + l_2$ using a Newton-Raphson algorithm summarizes the model used by McGilchrist & Aisbett (1991).

Defining $\mathbf{U} = \log(\mathbf{Z})$ and $\mathbf{U} \sim \mathrm{N}(\mathbf{0}, \sigma^2(\mathbf{I} - M^{-1}\mathbf{1}\mathbf{1}'))$ (or equivalently, $\mathbf{Z}$ follows a log-normal distribution), l_2 is the log density of $\mathbf{U}$. That is to say, if proposed steps of $\mathbf{U}$ are very low density, there will be little contribution from l_2 to the overall likelihood. Its variance matrix is singular, chosen to ensure the log frailties sum to 0; $\mathbf{1}'\mathbf{U} = 0$, as they are a relative measure. l_1 is a Taylor approximation (to the second degree) of l at the current location $\hat{\boldsymbol{\beta}}$ and $\hat{\mathbf{U}}$ of the Newton-Raphson algorithm. The combination of l_1 and l_2 results in a ridge type penalty term $+\sigma^{-2}\mathbf{I}$ being added to the frailties component in the hessian of l. It stabilizes the Newton-Raphson algorithm as frailties are restricted from growing uncontrollably.

The estimation procedure initializes values for all of $\boldsymbol{\beta}, \mathbf{U}$, and σ^2 (which determines the size of the regularization on frailties). The hessian is calculated at these values, and used to determine the next step for the Newton-Raphson algorithm, and the process repeats until full convergence.

3.2 Replicated Stan Model

A brief justification for the logic underlying the Bayesian replication methods follows. Further details on the derivations of both methods, as well as the Stan implementation are included in the appendix section B.

McGilchrist & Aisbett (1991) estimates their coefficients by maximizing the penalized partial log-likelihood $l = l_1 + l_2$ shown above. To transition away from the MLE (Frequentist) principles, 2 different approaches are implemented - an identical Newton-Raphson optimization approach to find the MAP of the posterior, and a Bayesian MCMC Sampling approach. The Stan model uses the same form where l_1 is viewed as the likelihood distribution $p(\mathbf{t}|\boldsymbol{\beta}, \mathbf{Z})$ and l_2 represents the prior distribution of u, $U \sim \text{Normal}(0, \sigma^2\mathbf{I})$, with $\sigma = 1$ fixed at the paper's value throughout, and β follows a uniform prior $p(\beta) \propto 1$. The normal prior on u is therefore not a weakly informative choice but the frailty distribution itself. The derived quantity $\sigma^2 = \mathbf{u'u}/(\mathbf{M} - 1)$ is also reported. The Stan implementation encodes this objective directly. The parameters are β and the vector $\boldsymbol{u}$ of 38 log-frailties, declared with a sum-to-zero constraint $\mathbf{1'u} = 0$ to match the identifiability restriction in the paper. Recurrence times are ordered smallest to largest, but not that they're distinct. McGilchrist & Aisbett (1991)'s model does not consider recurrence time ties Therneau (2026) resulting in bias McGilchrist & Aisbett (1991).

3.2.1 From penalized likelihood to log-posterior

$l = l_1 + l_2$ can be viewed with a Bayesian perspective where l_1 is the log-likelihood $\log p(\mathbf{t} \mid \boldsymbol{\beta}, \mathbf{u})$ and l_2 is the log-prior $\log p(\mathbf{u} \mid \sigma)$ with $\mathbf{u} \sim \mathcal{N}(\mathbf{0}, \sigma^2\mathbf{I})$. With the flat uniform prior $p(\boldsymbol{\beta}) \propto 1$, the log-posterior is

$$\log p(\boldsymbol{\beta}, \mathbf{u} \mid \mathbf{t}) = l_1(\boldsymbol{\beta}, \mathbf{u}) + l_2(\mathbf{u}) + \text{const} \tag{3}$$

The Stan implementation mirrors this, by further simplifying the likelihood. We use the transformation

$$\eta_{ij} = \mathbf{x}_{ij}\boldsymbol{\beta} + U_i \tag{4}$$

where i is the patient, and j is the interval that belongs to that patient.

The paper's partial likelihood (2) therefore reduces to the form, for $D_{in} = 1$:

$$l(\beta, U) = \sum_{n=1}^{N}\left(\log \sum_{j=r_{in}}^{m_i} \exp(\eta_{ij}) - \log \sum_{j \in R_n} \exp(\eta_j)\right) \tag{5}$$

where $n = 1, \ldots, N = 76$ intervals ordered by recurrence time. Therefore, the frailty enters through η, and the per-patient summation never has to be carried out explicitly.

3.2.2 FITTING

Two different approaches are taken to maximize the log-posterior l to obtain the parameter estimates:

(i) Stan's **Newton optimizer (MAP)** (via *rstan::optimizing* with *algorithm = "Newton"* and *hessian = TRUE*), which is numerically identical to the Newton-Raphson approach done by McGilchrist & Aisbett (1991) and

(ii) **MCMC Sampling (NUTS)** (using the NUTS variant of Hamiltonian Monte Carlo). Although this yields the posterior mean rather than the posterior mode, MCMC samples the full posterior distribution, offering deeper inferential details and better scalability to higher dimensions.

Since the frailties posterior is skewed, these 2 approaches may not yield identical results, so differences in results reflect a choice of estimand rather than failure to replicate. Due to the log-normal skew, we utilise the posterior median point estimate rather than the posterior mean in the NUTS sampling. Reporting both allows us to separate the question "does our model reproduce the paper?" from the question "what does a fully Bayesian treatment of the same model say?".

3.3 On Comparing Models

Regression coefficient estimates for covariates *Age, Sex* and disease types *GN, AN*, and *PKD* with respective standard errors from the models will be compared with the original paper. Frailty point estimates between the original paper and replication estimates will also be compared.

3.4 Deviations from the paper

Estimating the frailties and risk coefficients with MCMC sampling is one clear deviation from the original paper. In contrast to the Newton-Raphson algorithm used to maximize the penalized partial log-likelihood, the NUTS sampler takes estimated draws directly from its distribution. In comparison, the MAP method aims to replicate the methodology of the paper as closely as possible. As mentioned, McGilchrist & Aisbett (1991) update σ^2 as the sample variance of the log-frailties, $\tilde{\sigma}^2 = (M-1)^{-1}\mathbf{U}'\mathbf{U}$, initializing it as 1. The exact update and interaction of this value with the published estimates is not fully specified and we were unable to identify the exact implementation that reproduces the reported $\sigma^2 = 0.3821$. We have therefore assumed that $\tilde{\sigma}^2$ is held fixed at its initialised value, and computing $\tilde{\sigma}^2 = (M-1)^{-1}\mathbf{U}'\mathbf{U}$ only as a generated value of the log-frailties. The final implementation in the paper remains unknown. Common to both estimation methods are two further possible deviations. The paper uses a Taylor series approximation of l for l_1, whereas we use the full partial log-likelihood as seen in (5).

Further, McGilchrist & Aisbett (1991) assume there are no ties in recurrence times when constructing the partial likelihood. This in reality is not the case, and there are ties in the data. This influences who remains in the risk set at t_n. We take the dataset as it comes from `survival`, ordering by time, but results could be sensitive to permutations of observations within tied recurrence times.

4 Results

4.1 Model diagnostics

Newton Optimizer (MAP) The 42×42 Hessian over $(\boldsymbol{\beta}, \mathbf{u})$ is negative definite: all eigenvalues are strictly negative, so the solution is a strict local maximum of the penalized log-posterior rather than a saddle point, thereby confirming that the optimization method found the strict local mode as desired.

MCMC Sampling (NUTS) We ran four chains of 50,000 iterations (half warm-up, thinned by 4) giving 25,000 post warm-up draws. No divergent transitions occurred. The Rank Normalised $\hat{R}$ was below 1.001 for all 43 parameters. The effective sample size were well above 90% of the posterior draws suggesting that the MCMC chains mixed efficiently with low autocorrelation, yielding highly reliable posterior estimates.

Further information can be found in the Appendix, Tables 4 & 3

4.2 Results

Frailties. Replicated frailty estimates are plotted against the paper's frailty estimates, with perfect replication sitting on the diagonal line. A full comparison of frailty results are

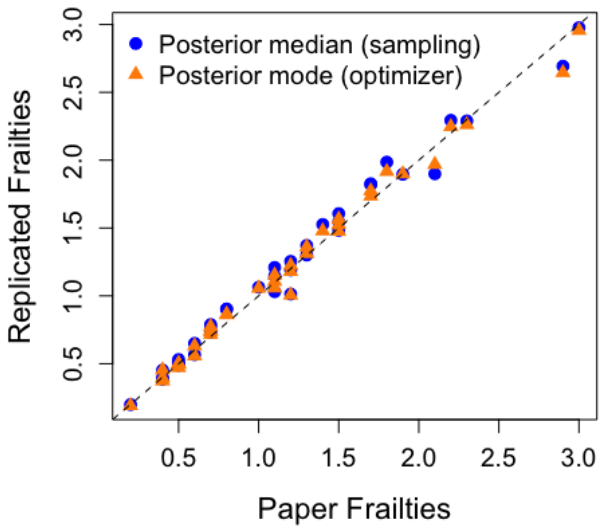


Figure 1. Frailty Estimates Comparison

The frailty estimates broadly align on visual inspection. Neither estimation method shows a clear advantage: the

MAP and NUTS estimates alternate in their proximity to the target diagonal, with no consistent pattern favouring one over the other.

Risk Coefficients & Variance of log-frailties. The data shown in Table 1 shows the results of the $\boldsymbol{\beta}$ estimates as well as the generated quantity, σ^2, representing the strength of the regularisation on the frailties. These values are presented alongside their standard errors. The number of decimal places presented are chosen to match the precision given in the paper.

Parameter	Paper	MAP	NUTS
$\hat{\beta}_{age}$	0.0063 (0.0134)	0.0055 (0.0179)	0.0059 (0.0184)
$\hat{\beta}_{sex}$	-1.7947 (0.4337)	-1.8502 (0.5374)	-1.9203 (0.5540)
$\hat{\beta}_{GN}$	0.2062 (0.4840)	0.2227 (0.6443)	0.2137 (0.6638)
$\hat{\beta}_{AN}$	0.4099 (0.4937)	0.4238 (0.6492)	0.4362 (0.6648)
$\hat{\beta}_{PKD}$	-1.2961 (0.7120)	-1.0792 (0.9346)	-1.2043 (0.9646)
$\hat{\sigma}^2$	0.3821	0.3921	0.3861

Table 1. $\hat{\boldsymbol{\beta}}$ estimates with respective standard errors, and estimated variance of the log frailty estimates $\hat{\sigma}^2$

The two estimation methods alternate at being closer to the paper's estimates, with both either over- or under-estimating each paper's coefficient. There is no consistent pattern of the replication estimates being particularly 'more accurate'. For example, our estimates for $\hat{\beta}_{AN}$ are just as far away from the paper values as $\hat{\beta}_{sex}$. The comparatively poorest result was the MAP estimate for $\hat{\beta}_{PKD}$. Further discussion of these discrepancies are given in the conclusions. It is also noticeable that every standard error and the variance of the frailties are greater in our replication estimates than the original paper.

5 Conclusions

5.1 Discussion

Overall, there remains more to understand about the paper to replicate the paper's estimates even closer.

For the MAP estimates, it was thought that the implementation was as close a replication as possible, and that maximizing the penalized log-likelihood with a Newton-Raphson algorithm would yield the same estimates. We tested some changes, in the search for improvements. As mentioned in section 3, σ^2 updates are slightly ambiguous in the paper. Where we have assumed $\sigma^2 = 1$, only updating it as the sample variance of the log-frailties after they have converged, this estimation procedure could also be interpreted as updating iteratively after each new log-frailty step. We attempted incorporating the latter method in R, but found negligible improvements. We separately attempted using a $N(0, 1)$ weakly informative prior on $\boldsymbol{\beta}$, but found no effect on the resulting estimates. Different orderings for tied recurrence times were tested, but very little change occurred to the estimates.

For the NUTS estimates, some deviations from the paper were expected simply due to the differing estimation approaches; MLE vs Bayesian sampling. One noticeable feature of the NUTS estimates were the greater standard errors for every $\hat{\beta}$ estimate. It is suspected that these standard errors are actually more accurate, given the MAP standard errors are estimates derived from the asymptotic covariance matrix, and are possibly underestimated given the relatively small sample size.

The relative difficulty of estimating $\hat{\beta}_{PKD}$ could be due to the fact there are only 6 uncensored observations of patients with the disease. Due to a relatively small dataset in general, estimations are sensitive to outliers. An interesting point of influence is Patient 21 due to his high recurrence times, who Therneau (2026) note that, when removed, collapses the need for estimating frailties at all in this dataset. A general criticism of the estimates found by McGilchrist & Aisbett (1991) are the lack of dealing with tied recurrence times, but the method of introducing log-normally distributed frailties to form a ridge type penalty is regarded as revolutionary.

5.2 Potential Improvements

Further investigation into σ^2 updating and estimation, we believe, is the biggest area for further potential improvements. Although attempting this briefly ourselves, a deeper dive into it is definitely warranted. McGilchrist (1993) additionally introduced a considerable improvement on the estimation method covered in this paper, it would be interesting to investigate those changes made in further detail. Finally, comparisons of this analysis against frequentist techniques that do include methods for tie handling (like Breslow or Efron) and comparing Stan implemented analyses with these would be another study of interest.

6 Acknowledgments

We hereby grant our consent for the utilization of this project report as a reference material within the context of future editions of the course.

References

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Therneau, T. M. *A Package for Survival Analysis in R*, 2026. URL https://cran.r-project.org/web/packages/survival/survival.pdf. R package version 3.8-12.

A Posteriordb

A formal submission, through a pull request, will be made to posteriorDB containing all required objects for a valid submission.

B Stan Code

```
data {
  int<lower=1> M, N, P;                    // # of patients, total obs, # of covariates
  vector[N] t;                             // gap times, sorted ascending
  array[N] int<lower=0, upper=1> D;        // 1 = infection, 0 = censored
  array[N] int<lower=1, upper=M> pid;      // patient_id for each interval
  matrix[N, P] X;                          // covariates- age,sex,disease type
  real<lower=0> sigma;                     // fixed log-frailty sd
}
parameters {
  vector[P] beta;                          // regression coeffs
  sum_to_zero_vector[M] u;                 // log-frailties restriction [1'u = 0]
}
transformed parameters {
  vector[N] eta = X * beta + u[pid];       // linear substitution
}
model {
  // Priors
  u ~ normal(0, sigma);                    // log-frailties ~ normal(0, sigma^2)

  // Cox's partial likelihood: numerator sums the patient's intervals still at risk
  for (k in 1:N)
    if (D[k] == 1) {
      real num = negative_infinity();
      for (j in k:N)
        if (pid[j] == pid[k])
          num = log_sum_exp(num, eta[j]);
      target += num - log_sum_exp(eta[k:N]);
    }
}
generated quantities {
  vector[M] Z = exp(u);                    // Frailties
  real sigma2 = dot_self(u) / (M - 1);     // sigma^2 = u'u/(M-1)
}
```

C A derivation of Cox's (1975) partial likelihood

The derivation begins with a definition for the probability of an event occurring to a given individual i at time point t_n, defined as their proportion of the total hazard of all individuals in the risk set (R) at t_n.

$$Pr(i \text{ has the event} | \text{an event occurs at } t_n) = \frac{h(\mathbf{x}_{in}, t_n)}{\sum_{m \in R} h(\mathbf{x}_{mt}, t_n)}$$

Substituting in Cox's (1972) proportional hazard model with frailty, we can complete the derivation. In the following r_{in} is defined by McGilchrist & Aisbett (1991) as 1 + the number of events or censoring times occurring to individual i before time t_n, where m_i is the total number of recurrences recorded on individual i. When $r_{in} > m_i$, $\alpha_{in}(\boldsymbol{\beta}) = 0$. $\alpha_{in}(\boldsymbol{\beta})$ then in one sentence is measure of a person's total systematic risk (ignoring the baseline) at t_n.

$$\begin{aligned}
\frac{h(\mathbf{x}_{in}, t_n)}{\sum_{m \in R} h(\mathbf{x}_{mt}, t_n)} &= \frac{Z_i \lambda(t_n) \alpha_{in}(\boldsymbol{\beta})}{\sum_{m=1}^{M} Z_m \lambda(t_n) \alpha_{mn}(\boldsymbol{\beta})} && \text{Where: } \alpha_{in}(\boldsymbol{\beta}) = \sum_{j=r_{in}}^{m_i} \exp(\mathbf{x}'_{ij}\boldsymbol{\beta}) \\
&= \frac{Z_i \alpha_{in}(\boldsymbol{\beta})}{\sum_{m=1}^{M} Z_m \alpha_{mn}(\boldsymbol{\beta})} && \text{(The baseline hazard cancels, it is a partial likelihood)} \\
&= Z_i b_{in}(\boldsymbol{\beta}, \mathbf{Z}) && \text{Where: } b_{in}(\boldsymbol{\beta}, \mathbf{Z}) = \frac{\alpha_{in}(\boldsymbol{\beta})}{\sum_{m=1}^{M} Z_m \alpha_{mn}(\boldsymbol{\beta})}
\end{aligned}$$

Moving to the joint likelihood, the dummy variable D_{in} is incorporated to only include observations that were observed on individual i at time point t_n (so that $D_{in} = 1$). Let $\mathbf{t}$ denote the vector of observed recurrence times.

$$
\begin{aligned}
L(\boldsymbol{\beta}, \mathbf{Z}|\mathbf{t}) &= \prod_{i=1}^{M} \prod_{n=1}^{N} \left(Z_i b_{in}(\boldsymbol{\beta}, \mathbf{Z})\right)^{D_{in}} \\
l(\boldsymbol{\beta}, \mathbf{Z}|\mathbf{t}) &= \log \prod_{i=1}^{M} \prod_{n=1}^{N} \left(Z_i b_{in}(\boldsymbol{\beta}, \mathbf{Z})\right)^{D_{in}} && \text{(Going to the partial log-likelihood)} \\
&= \sum_{i=1}^{M} \sum_{n=1}^{N} D_{in} \log\left(Z_i b_{in}(\boldsymbol{\beta}, \mathbf{Z})\right) && \text{(Simplifying)}
\end{aligned}
$$

Which is the result utilized by McGilchrist & Aisbett (1991).

D Further simplifications for Stan

As established above, McGilchrist & Aisbett (1991) give the log partial likelihood as

$$
l(\boldsymbol{\beta}, \mathbf{Z}|\mathbf{t}) = \sum_{n=1}^{N} \sum_{i=1}^{M} D_{in} \, \ln(Z_i \, b_{in}) \,, \qquad b_{in} = \frac{\alpha_{in}(\boldsymbol{\beta})}{\sum_{m=1}^{M} Z_m \, \alpha_{mn}(\boldsymbol{\beta})}, \qquad \alpha_{in}(\boldsymbol{\beta}) = \sum_{j=r_{in}}^{m_i} g(\mathbf{x}_{ij})
$$

where α_{in} sums the risk of patient i's intervals still at risk at t_n, and $Z_i b_{in}$ is the probability that it is individual i who fails at t_n, given a failure at t_n.
Expanding b_{in}, and substituting $g(\mathbf{x}) = \exp(\mathbf{x}\boldsymbol{\beta})$ and $Z_i = e^{U_i}$:

$$
l(\beta, U) = \sum_{n=1}^{N} \sum_{i=1}^{M} D_{in} \left(U_i + \log \sum_{j=r_{in}}^{m_i} \exp(\mathbf{x}_{ij}\boldsymbol{\beta}) - \log \sum_{m=1}^{M} \sum_{j=r_{mn}}^{m_m} \exp\left(U_m + \mathbf{x}_{mj}\boldsymbol{\beta}\right) \right)
$$

Let $\eta_{ij} = \mathbf{x}_{ij}\boldsymbol{\beta} + U_i$. We can simplify the second double summation $\{i, j\}$ that counts every interval still at risk at t_n, i.e. every interval with time $t \geq t_n$. This is denoted as R_n, the risk set:

$$
\begin{aligned}
l(\beta, U) &= \sum_{n=1}^{N} \sum_{i=1}^{M} D_{in} \left(\log[\exp(U_i)] + \log \sum_{j=r_{in}}^{m_i} \exp(\mathbf{x}_{ij}\boldsymbol{\beta}) - \log \sum_{j \in R_n} \exp(\eta_{j,n}) \right) \\
l(\beta, U) &= \sum_{n=1}^{N} \sum_{i=1}^{M} D_{in} \left(\log \sum_{j=r_{in}}^{m_i} \exp(\eta_{ij}) - \log \sum_{j \in R_n} \exp(\eta_{j,n}) \right)
\end{aligned}
$$

In the paper, McGilchrist & Aisbett (1991) define D_{in} as a dummy variable that = 1 if an **uncensored** recurrence occurs to person i and time point t_n. We can therefore see in the above, that a recurrence only contributes to the likelihood if it is uncensored. Given a recurrence is assumed to only happen to one individual at t_n, we can drop the summation over i, which is only applicable if event times were not distinct.
Considering only uncensored recurrences $D_{in} = 1$, the summation over all observations N includes only n's where this is true. The personal hazard (left term of the above) calculates the total hazard for i at t_n for the uncensored event. By separating the notation of the risk set to depend only on n, the risk set is ensured to include all observations, both censored and uncensored, that occurred at or after t_n.

$$
l(\beta, U) = \sum_{n=1}^{N} \left(\log \sum_{j=r_{in}}^{m_i} \exp(\eta_{ij}) - \log \sum_{j \in R_n} \exp(\eta_j) \right) \qquad \text{Where: } D_{in} = 1
$$

This simplified likelihood expression forms the basis of what was built and modelled in Stan.

E Full Results Table

Patient	Paper	MAP	NUTS	Patient	Paper	MAP	NUTS
1	2.30	2.26	2.29	20	1.30	1.31	1.30
2	1.90	1.90	1.90	21	0.20	0.19	0.20
3	1.20	1.22	1.25	22	0.60	0.56	0.57
4	0.50	0.50	0.53	23	1.70	1.77	1.83
5	1.50	1.48	1.48	24	1.00	1.06	1.06
6	1.10	1.15	1.21	25	0.70	0.72	0.73
7	3.00	2.96	2.98	26	0.50	0.50	0.51
8	0.50	0.47	0.49	27	1.10	1.11	1.15
9	0.70	0.74	0.77	28	1.80	1.92	1.99
10	0.40	0.37	0.39	29	1.50	1.53	1.55
11	0.60	0.63	0.65	30	1.50	1.57	1.61
12	1.20	1.18	1.19	31	1.70	1.74	1.82
13	1.40	1.48	1.53	32	1.30	1.36	1.37
14	0.40	0.45	0.45	33	2.90	2.65	2.69
15	0.40	0.40	0.40	34	0.70	0.77	0.79
16	1.10	1.06	1.03	35	2.20	2.25	2.29
17	0.80	0.87	0.90	36	0.70	0.74	0.73
18	0.80	0.86	0.90	37	2.10	1.97	1.90
19	0.50	0.50	0.49	38	1.20	1.01	1.01

Table 2. Estimated frailties $Z_i = \exp(u_i)$ for each of the 38 patients: the values reported in Table 1 of McGilchrist and Aisbett (1991), the posterior mode from the optimizer (MAP), and the posterior median from NUTS sampling.

Parameter	$\hat{R}$	n_{eff}
$\hat{\beta}_{age}$	1.000	24051
$\hat{\beta}_{sex}$	1.000	23858
$\hat{\beta}_{GN}$	1.000	24524
$\hat{\beta}_{AN}$	1.000	24051
$\hat{\beta}_{PKD}$	1.000	24395
$Z_1, \ldots, Z_{38}$ (range)	1.000–1.000	22648–24542

Table 3. Convergence diagnostics for the NUTS fit. $\hat{R}$ is the split-chain potential scale reduction factor and n_{eff} the effective sample size; the frailty row gives the range across all 38 patients. All $\hat{R}$ values are below 1.01 and all effective sample sizes exceed 20,000, indicating well-mixed chains.

Diagnostic	Value
Post-warmup draws (4 chains)	25000
Divergent transitions	0
Optimizer return code	0
Largest Hessian eigenvalue	-0.525
Smallest Hessian eigenvalue	-13245.237

Table 4. Fit-level diagnostics. No divergent transitions occurred during sampling. For the optimizer, a return code of 0 indicates successful convergence, and a strictly negative largest eigenvalue of the Hessian confirms that the solution is a strict local maximum of the posterior.